SHAWN RAO

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SUMMARY

Data Science undergraduate in the University of Michigan College of Engineering, graduating December 2027, applying for the Summer 2027 Software Engineering Internship. Builds responsive web interfaces in JavaScript, HTML5 and CSS3 and performance-sensitive systems in C++, working in Git-based collaborative development workflows.

EDUCATION

University of Michigan, College of Engineering Ann Arbor, MI
Bachelor of Science in Engineering (B.S.E.), Data Science *Expected 12/2027*

Relevant Coursework: SI 339 (Web Design, Development & Accessibility), SI 206 (Data-Oriented Programming), EECS 281 (Data Structures & Algorithms), EECS 280, EECS 203 (Discrete Math), EECS 376 (Foundations of CS), ROB 101 (Computational Linear Algebra), MATH 215, STATS 250

In Progress (Fall 2026): DATASCI 306 (Statistical Computing in R), STATS 412 (Probability & Statistics), EECS 201 (Unix, Shell, Version Control)

Certificate: Applied Generative AI Specialization, Purdue University Online *2025*
Online, 16 weeks. LLM architecture, transformers and attention, prompt engineering, retrieval-augmented generation; LangChain, OpenAI API, Azure AI Studio.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), C++, Python, R, SQL, HTML5, CSS3
Front End: DOM manipulation, CSS Grid, Flexbox, media queries, responsive design, React, Node.js
Version Control: Git (interactive rebase, hooks, reflog recovery, conflict resolution), GitHub, GitLab, collaborative workflows
Systems & Shell: Linux, Unix, Bash, Zsh, shell scripting, process control (fork/exec/wait), grep, sed, POSIX regex, Docker, AWS EC2 and S3
Testing: Randomized differential testing, mutation testing, Valgrind, strict-flag builds, WCAG and keyboard audits
Data: Pandas, NumPy, tidyverse, SQL (window functions, recursive CTEs), LangChain, OpenAI API

PROJECTS

Conversational Search Interface *JavaScript, HTML5, CSS3, Python, LangChain, OpenAI API*

- Built a responsive web interface in JavaScript, HTML5 and CSS3, using CSS Grid and Flexbox for layout and media queries to hold the design together from mobile widths to desktop.
- Rendered streamed responses through direct DOM manipulation rather than re-rendering the page, and handled loading, empty and error states so the interface stayed usable when the backend was slow or returned nothing.
- Built the retrieval layer in Python over an embedded document index, returning answers with inline citations, then tuned chunk size and retrieval depth against a hand-labeled test set.

Order-Matching Engine and Market Simulator *C++, STL*

- Built a discrete-event engine that reads a stream of timestamped buy and sell orders and matches them against a price-time priority book, using paired binary heaps so each insert and match runs in $O(\log n)$.
- Produced running median price, per-trader transfer totals, and time-travel arbitrage reports in a single pass over the order stream, with no second read of the input.
- Analyzed the performance bottleneck and cut runtime on the largest test inputs by replacing linear scans with a custom priority queue, choosing the data structure for scalability rather than for the smallest change.

EXPERIENCE

University of Michigan, College of Engineering (CAEN) Ann Arbor, MI
Computer Hardware Technician, Student Staff *04/2023 – 04/2026*

- Supported 200+ Linux, Windows and macOS workstations and servers in shared engineering labs, holding 99%+ uptime for 500+ students and faculty.
- Diagnosed hardware, software and network failures to root cause under walk-up time pressure, then documented each fix so the rest of the team could repeat it.
- Reduced hardware downtime 25% by standardizing imaging and diagnostics.
- Supported a CrowdStrike endpoint rollout to full fleet coverage; integrated 3 new lab technologies, cutting setup time 40%.

NYX Inc. Livonia, MI
Information Technology Summer Intern *04/2021 – 07/2021*

- Configured and deployed employee workstations to company servers; tracked and reconciled IT hardware inventory.
- Supported office and plant staff on Windows desktops and on the production floor.